llmXive Automated Review of Multi-Turn Reflective Masking Elicits Reasoning in Mask Diffusion Models

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so errors, misreadings, and inaccuracies are likely. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the llmXive agent-prompt library; each section below also links the specific prompt it used.

This report is advisory, automated feedback for the authors. llmXive does not modify the paper, and nothing is accepted or rejected on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

The paper makes several strong claims regarding the capabilities of Reflective Masking (RM) and History Reference (HR) that are generally well-supported by the provided data, but a few specific assertions require nuance or clarification to ensure strict factual accuracy.

First, in Section 3.1, the authors state that the inference rule is "driven entirely by the model's per-position output distribution" (lines 135-136). However, Equation 1 explicitly defines the "Reveal" action as a deterministic `argmax` operation. While the "Re-mask" and "Keep" decisions rely on probability comparisons, the "Reveal" step discards the distribution's entropy in favor of a single token. This contradicts the claim of being "entirely" distribution-driven, as it introduces a deterministic bottleneck that prevents the model from sampling diverse corrections during the reveal phase. The text should clarify that the *decision* to reveal is distribution-driven, but the *selection* of the revealed token is deterministic.

Second, in Section 4.3, the authors claim that "MATH500 evaluation focuses primarily on the final answer" (lines 338-339) to explain why the performance gain is smaller than on MBPP. While the MATH500 metric is indeed final-answer accuracy, the paper's own methodology emphasizes the correction of "intermediate reasoning steps" (CoT). If the evaluation protocol does not penalize incorrect intermediate steps that lead to a correct final answer (or vice versa), the claim that the method's benefits are limited by the metric is valid. However, if the benchmark implicitly relies on step-wise correctness (e.g., via a verifier), the claim is an oversimplification. The text should explicitly state that the MATH500 benchmark used evaluates *only* the final answer, distinguishing it from benchmarks that might verify intermediate steps.

Third, in Section 4.3, the authors claim that "RM method benefits more in code tasks, where iterative revision can correct a greater number of token-level errors" (lines 342-344). This is a plausible hypothesis, but the paper does not provide direct evidence (e.g., a count of corrected tokens per task) to support the claim that code tasks inherently have "a greater number of token-level errors" than math tasks. The data shows a larger *performance gain* on MBPP, but attributing this solely to the *number* of errors without quantifying the error density or correction rate is a slight overreach. The text should qualify this as a "likely explanation" rather than a definitive fact.

Finally, in Section 4.2, the text states that "introducing a decay factor alone still improves over the no-history baseline" (lines 285-286). Table 2 shows the "decay" variant (89.4%) indeed outperforms the "no-history" baseline (82.4%). However, the text also claims this variant "performs worse than the \temporalmethod-only variant" (91.4%). This is accurate. The phrasing is slightly dense but factually correct. No revision is strictly required here, but the sentence structure could be improved for clarity.

Overall, the claims are largely supported, but the deterministic nature of the reveal step and the specific attribution of performance gains to error counts need slight qualification to avoid overstatement.
Action items.

- [writing] In Section 3.1, the claim that inference is 'driven entirely by the model's per-position output distribution' contradicts the deterministic argmax used for the 'Reveal' action in Eq. 1. Clarify that the decision to reveal is probabilistic, but the token selection is deterministic.
- [writing] In Section 4.3, the claim that MATH500 'focuses primarily on the final answer' is used to explain lower gains. Explicitly confirm the benchmark evaluates only the final answer, as the method targets intermediate CoT steps which might otherwise be penalized in other protocols.
- [writing] In Section 4.3, the attribution of larger MBPP gains to 'a greater number of token-level errors' is a hypothesis. The paper lacks data on error density or correction counts. Qualify this as a likely explanation rather than a proven fact.
- [writing] In Section 4.2, the text claims the decay variant 'improves over the no-history baseline' (89.4% vs 82.4%). While numerically true, the phrasing is dense. Ensure the comparison clearly distinguishes between the 'decay-only' and 'HR-only' ablations to avoid confusion.

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_figure_critic.md`

Figure 1

Figure 1 presents a grid of qualitative image editing results comparing different models. However, the visualization of the predicted masks (red overlays) and the specific column headers are not defined in the caption, requiring the reader to infer their meaning.

Figure 2

The figure presents qualitative image editing results with clear visual comparisons, but the caption contains a grammatical error and incorrectly describes the presence of heat maps that are not visible in the figure.

Figure 3

Figure 3 effectively illustrates the synthetic data construction process, but there is a slight disconnect between the 'Wrong' sequence rule definition and its application in the 'Current

Step' example, and the legend could more clearly link the 'Noisy' and 'Wrong' token states.

Figure 4

The figure content (math problem steps) contradicts the caption's description ('text generation'), and the figure lacks necessary context to interpret the displayed results.

Figure 5

The figure illustrates a reasoning process but suffers from a grammatically incomplete caption and confusing step numbering. Additionally, the mathematical content displayed in the steps does not align with standard algebraic procedures for the problem presented. Action items.

- [science] Figure 1: The 'Predict Mask(Ours)' row displays red overlays on the original images, but the caption does not define this visualization method or explain that red indicates the predicted mask area.
- [writing] Figure 1: The column headers in the top section (e.g., 'Predict Mask(Ours)', 'Ours', 'Lumina') are not explicitly defined in the caption, leaving the reader to infer that these represent different model outputs.
- [writing] Figure 2: The caption states 'The masks predicted by are highlighted in red,' which contains a grammatical error and missing subject (likely 'by our model').
- [writing] Figure 2: The caption claims 'Heat maps at the bottom visualize pixel-wise differences,' but the figure displays a grid of qualitative image editing results without any heat maps.
- [science] Figure 2: The column labeled 'Predict Mask(Ours)' shows red overlays, but the caption's claim about heat maps visualizing pixel-wise differences is not supported by the visual content shown.
- [science] Figure 3: The 'History construction rules' panel shows a 'Wrong' sequence starting with w0, but the diagram below it shows the 'Current Step' sequence starting with '2' (a correct token), creating a disconnect between the rule definition and the application example.
- [writing] Figure 3: The legend at the bottom left defines 'Noisy' as a dashed box, but the 'Sample wrong' section uses solid orange boxes for wrong tokens without explicitly linking them to the 'Wrong' legend entry (solid orange box) in a way that clarifies the transition from 'Noisy' to 'Wrong'.
- [science] Figure 4: The figure displays mathematical problem-solving steps (Cases 1-5) rather than 'text generation' results, contradicting the caption's claim.
- [science] Figure 4: The figure lacks a descriptive caption explaining the specific tasks, the model's performance, or the meaning of the red/green markers, making it impossible to interpret the 'qualitative results'.

- [writing] Figure 5: The caption begins with a lowercase verb ('actively re-masks...') and lacks a subject, failing to identify the model or method being illustrated.
- [science] Figure 5: The mathematical problem shown at the top is a linear inequality, but the solution steps display 'M' tokens and arithmetic operations that do not correspond to standard algebraic solving methods for inequalities.
- [writing] Figure 5: The 'Step' labels (45, 115, 135, 138, 139) are non-sequential and arbitrary, making it difficult to follow the chronological progression of the reasoning process.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_jargon_police.md`

The manuscript relies heavily on field-specific acronyms and shorthand that are not consistently defined at their first occurrence, creating barriers for non-specialist readers. In the Abstract, 'MDM' and 'AR' are used immediately without expansion. While 'Mask Diffusion Models' and 'autoregressive' appear in the Introduction, the Abstract should be self-contained. Similarly, 'SFT' is used in Section 5.1 without definition, and 'CoT' appears in Section 5.3 without the full phrase 'Chain-of-Thought' preceding the acronym.

In the appendices, the density of undefined jargon increases. 'VQ tokens' (Appendix A.4), 'CFG dropout' (Appendix A.5), and 'MAE-RGB' (Section 5.1) are presented as if they are common knowledge. Specifically, 'CFG' is a critical mechanism in diffusion models, but its full name is never spelled out in the text. The use of 'w.r.t.' in the preamble shims (though not in the main text) and the heavy reliance on mathematical shorthand like 'TV' (Total Variation) without a brief parenthetical explanation in the main text (it is defined in the preamble but not the prose) further alienates general readers.

To improve accessibility, the authors should ensure every acronym is expanded upon its first appearance in the main text (Abstract and Introduction are critical). Terms like 'SFT', 'CoT', 'CFG', and 'VQ' should be written out fully before being abbreviated. Additionally, specific metric variants like 'MAE-RGB' should be briefly described to clarify their specific calculation method. These changes would significantly lower the entry barrier for readers outside the immediate sub-field of diffusion language models.
Action items.

- [writing] The manuscript relies heavily on field-specific acronyms and shorthand that are not consistently defined at their first occurrence, creating barriers for non-specialist readers. In the Abstract, 'MDM' and 'AR' are used immediately without expansion. While 'Mask Diffusion Models' and 'autoregressive' appear in the Introduction, the Abstract should be self-contained. Similarly, 'SFT' is used in Section 5.1 without definition, and

'CoT' appears in Section 5.3 without the full phrase 'Chain-of-Thoug

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_logical_consistency.md`

The paper presents a coherent high-level argument: Mask Diffusion Models (MDMs) support local edits, but standard decoding is passive; therefore, training the model to actively decide when to re-mask (Reflective Masking) should enable reasoning. However, several logical gaps exist where conclusions do not strictly follow from premises or rely on unverified assumptions.

First, regarding the inference mechanism (Section 3.1), the authors define a deterministic rule (Eq. 1) that can lead to state loops. They introduce History Reference (Section 3.2) to break these loops by altering the input conditioning. While the intuition is plausible, the paper asserts the rule is "unchanged" while the input changes, and that this "addresses" the loop. Logically, changing the input distribution does not guarantee the elimination of fixed points in a deterministic system unless the new dynamics are proven to be non-cyclic or contractive. The paper lacks a formal argument or empirical evidence (e.g., trajectory length analysis) proving that History Reference *necessarily* breaks the loops rather than just shifting them. The conclusion that the method "enables test-time scaling" relies on the unproven assumption that the process terminates.

Second, the theoretical justification in Appendix A.1 (Theorem 1) relies on a "rich-family idealization" where the model class can perfectly represent the conditional label distribution. The paper claims minimizing the training loss drives the model toward this distribution. However, the experiments use specific, finite-capacity models (e.g., a 0.81M parameter model for Sudoku). The logical leap from the population minimizer (infinite capacity) to the empirical performance of these specific small models is not bridged. The paper does not discuss approximation error or conditions under which the finite model class can approximate the oracle well enough to justify the "Bayes-optimal" claims. The conclusion that the method "activates" a latent capability assumes the model has the capacity to learn it, which is an empirical claim not fully supported by the theoretical premises.

Finally, in the Sudoku ablation study (Section 4.2), the logical distinction between the "History Reference" mechanism and the "decay" parameter is ambiguous. Equation 2 defines the History Reference embedding explicitly including a decay factor γ. The table compares a variant without decay (implied) against one with decay. The text claims "introducing a decay factor alone... performs worse," implying decay is a separate component. However, if the mechanism *is* the accumulation with decay, then the "no decay" variant is a specific instantiation, not a separate one. The causal claim that "properly structuring historical information" is crucial is sound, but the specific attribution of gains

to “decay” versus “rotation” is muddied by the definition of the mechanism itself. The paper should clarify whether “History Reference” refers to the accumulation structure (with or without decay) or specifically the decayed accumulation to ensure logical consistency in the ablation.
Action items.

- [science] Section 3.1 claims the deterministic inference rule causes loops, and Section 3.2 asserts History Reference fixes this by changing input. However, the paper does not logically prove that the new input distribution breaks the fixed-point cycle, only that it ‘addresses’ it. A formal argument or empirical proof of termination is missing.
- [science] Appendix A.1 (Theorem 1) assumes a ‘rich-family’ model class to prove convergence to the oracle distribution. The experiments use finite-capacity models (e.g., 0.81M params). The paper fails to logically bridge the gap between the infinite-capacity theoretical guarantee and the empirical performance of the specific small models used.
- [writing] In Section 4.2, the ablation distinguishes between ‘History Reference’ and ‘decay’, yet Eq. 2 defines the mechanism inherently with decay. The claim that ‘decay alone’ performs worse is logically ambiguous because the mechanism definition includes decay. The distinction between the mechanism structure and the decay parameter needs clarification.

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_overreach.md`

The paper makes several claims that extend beyond the provided empirical evidence, particularly regarding the nature of “reasoning” and “test-time scaling.”

First, the central claim that the method “elicits reasoning” (Title, Abstract, Introduction) is an over-interpretation of the results. The training paradigm (Section 3.3, Eq. 3) relies on an oracle revision rule $\tau(z_i, \xstar_i)$ that requires access to the ground-truth target $\xstar_i$ to determine the correct action (mask, reveal, or keep). The model is trained to mimic this oracle on synthetic trajectories. While the method improves performance on benchmarks like MATH500 and MBPP, the gains are modest (e.g., −2.4% on MATH500, Table 2) and the model is essentially learning to correct errors it was explicitly taught to identify during training. This demonstrates the successful transfer of a correction policy, but does not substantiate the claim that the model has acquired a general “reasoning” capability or can self-correct in the absence of such strong supervisory signals during training. The term “reasoning” should be tempered to “iterative refinement” or “error correction” unless the authors can demonstrate emergent reasoning on out-of-distribution tasks without oracle guidance.

Second, the assertion that the approach provides "native test-time scaling" (Abstract, Introduction) is not fully supported. The paper fixes the number of denoising steps to $T=6$ for all experiments (Appendix A.3). To justify the term "scaling," the authors should demonstrate that increasing the number of inference steps (and thus compute) leads to consistent performance improvements, ideally showing a scaling law or convergence behavior that outperforms baselines under increased compute budgets. Without such analysis, the claim remains speculative.

Finally, the description of History Reference as "parameter-free" (Abstract, Introduction) is technically accurate regarding learnable weights but potentially misleading regarding computational cost. The method requires maintaining a running accumulated embedding $a_i^{(t)}$ for every token at every step, involving matrix rotations and additions. This introduces non-trivial memory and compute overhead compared to standard MDM inference. The paper should clarify this trade-off rather than implying the mechanism comes at no cost.

Action items.

- [science] The claim that RM provides 'native test-time scaling' (Abstract, Intro) is overreaching. The paper demonstrates performance gains with fixed T=6 steps (Appendix A.3) but does not empirically show that increasing T (scaling compute) yields monotonic improvements or converges to a better solution compared to baselines. This needs explicit scaling curves to justify the 'scaling' terminology.
- [writing] The assertion that the method 'elicits reasoning' (Title, Abstract) is too strong for the Sudoku and text results. The improvements on MATH500 are marginal (~2.4%) and the method relies on oracle-labeled synthetic data (Eq. 3) that explicitly teaches the model to correct known errors. This demonstrates learning a correction policy, not necessarily emergent reasoning capabilities beyond the training distribution.
- [writing] The claim that History Reference is 'parameter-free' (Abstract, Intro) is technically true regarding learnable weights but overstates the efficiency. The method requires maintaining and updating an accumulated embedding vector $a_i^{(t)}$ for every position at every step, which increases memory bandwidth and compute overhead compared to standard MDM inference. The paper should clarify this trade-off rather than implying zero cost.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_safety_ethics.md`

The manuscript presents a novel method for enabling reflective reasoning in Mask Diffusion Models (MDMs) through a lightweight post-training paradigm. From a safety and ethics perspective, the work does not raise immediate red flags regarding dual-use risks, data privacy violations, or the generation of harmful content.

The training methodology relies on synthetic data construction (Section 3.3, Fig. 2) where “wrong tokens” are sampled from a corruption distribution ν. The authors explicitly state in Appendix A.5 that for text generation, this distribution is derived from a frozen, pre-trained checkpoint’s top-k predictions, excluding the ground truth. This approach avoids the need to scrape or utilize private, sensitive, or potentially harmful datasets for the specific purpose of training the revision mechanism. The use of standard, public benchmarks (MATH, MBPP, ARC-Challenge, ImgEdit) for evaluation further mitigates concerns regarding data privacy and consent, as these datasets are widely accepted in the research community and typically have established usage licenses.

The proposed “Reflective Masking” mechanism allows models to iteratively revise their own outputs. While this capability could theoretically be used to refine harmful outputs (e.g., making a generated hate speech more coherent or a malicious code snippet more functional), the paper frames this as a general reasoning primitive for error correction in tasks like math and code synthesis. The authors do not claim to optimize for adversarial robustness or jailbreak resistance, nor do they demonstrate the method’s ability to bypass safety filters. The “Limitations” section (Section 6) appropriately acknowledges that the current base models have limited reasoning capabilities, suggesting that the method is not yet a powerful tool for sophisticated adversarial attacks.

There are no indications of conflicts of interest, and the authors do not appear to be using human subjects in a way that would require IRB approval, as the evaluation is purely algorithmic on public benchmarks. The “User Study” mentioned in Table 1 (Image Editing) involves 29 participants evaluating image quality; while the paper does not detail the IRB approval for this specific study, it is a standard practice in computer vision to use small-scale user studies for preference ranking without full IRB oversight, provided no sensitive data is collected. Given the context of a preprint on a public archive, this is acceptable, though a brief statement on ethical approval for the user study in the final version would be a minor best practice.

Overall, the paper adheres to standard ethical guidelines for AI research. The method is a technical improvement on model architecture and training dynamics rather than a tool designed to exploit or harm. No action items are required from a safety and ethics standpoint.

Scientific Evidence

Weighs the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_scientific_evidence.md`

The scientific evidence supporting the claims of Reflective Masking (RM) and History

Reference (HR) is generally robust, particularly in the controlled Sudoku setting where the ablation study (Table 2, lines 330-345) clearly isolates the contribution of the history mechanism. The synthetic data construction (Fig. 2, lines 200-220) is well-defined, and the theoretical bounds in the appendix (Appendix A) provide a solid foundation for the training objective.

However, the evidence for the magnitude of gains in open-ended text generation requires stronger statistical validation. In Table 3 (lines 380-390), the improvement on MATH500 is modest (~2.4%). Without reported standard errors, confidence intervals, or p-values derived from multiple evaluation seeds, it is difficult to rule out that this gain is a result of variance in the benchmark split rather than a consistent model improvement. Similarly, the large gap between MBPP (~8.8%) and MATH500 gains is explained qualitatively but lacks a quantitative breakdown of error types to substantiate the "token count" hypothesis.

Furthermore, the user study for image editing (Table 1, line 275) relies on 29 participants but omits critical methodological details. To ensure the 14.9-point gain is not an artifact of presentation order or specific prompt selection, the review requires a description of the blinding procedure, randomization strategy, and the number of distinct image prompts evaluated per participant. Finally, the Sudoku ablation showing a drop in performance when adding decay without HER (Table 2) is interpreted as a structural failure of the decay mechanism alone. A sensitivity analysis over the decay hyperparameter γ for this specific variant is necessary to confirm the drop is not merely due to a suboptimal learning rate or decay value.

Action items.

- [science] The Sudoku ablation (Table 2) shows a performance drop when adding decay without HER (89.4% vs 91.4%). The text attributes this to 'insufficient' signal weakening, but does not rule out that the decay hyperparameter was simply suboptimal for that specific configuration. Report a sensitivity analysis or grid search over decay factors for the 'HR~decay' variant to confirm the drop is structural, not tuning-related.
- [science] The text generation results (Table 3) show a ~8.8% gain on MBPP but only ~2.4% on MATH500. The authors attribute this to token count differences, but do not provide statistical significance testing (e.g., bootstrap confidence intervals) to confirm these gains are not due to variance in the evaluation set, especially given the small delta on MATH500.
- [science] The image editing user study (Table 1) reports a score of 68.2 vs 53.3 for the baseline with 29 participants. The paper lacks details on the study protocol (e.g., randomization, blinding, number of samples per participant) required to assess the robustness of this statistical claim against selection bias or order effects.

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_statistical_analysis.md`

The statistical analysis presented in the manuscript is generally sound in its design, utilizing appropriate metrics for the specific tasks (e.g., Exact Accuracy for Sudoku, VQAScore for image editing). The theoretical appendix provides a rigorous derivation of the Bayes-consistency of the training objective (Theorem 1) and excess risk bounds (Theorem 2), which correctly frames the learning problem. However, the empirical validation of the proposed method's superiority lacks necessary statistical rigor in the reporting of results.

Specifically, in Section 4.1, the user study results (Table 1) are presented as point estimates (e.g., 68.2 vs. 53.3) without any measure of uncertainty. With a sample size of 29 participants, it is essential to report 95% confidence intervals for the mean preference scores and explicitly state the statistical test used (e.g., paired t-test or non-parametric equivalent) along with the resulting p-values to validate that the observed difference is statistically significant.

Similarly, in the ablation studies for Sudoku (Section 4.2, Table 2) and the main text generation benchmarks (Section 4.3, Tables 3 and 4), the reported improvements are presented as deterministic gains. The manuscript does not specify the number of random seeds used for training or evaluation, nor does it provide standard deviations or confidence intervals. Without this information, it is impossible to distinguish between genuine methodological improvements and random variance, particularly for the smaller gains observed in the MATH500 subset (e.g., −0.59% in Algebra). The authors should re-run experiments with multiple seeds (typically 3-5) and report the mean ± standard deviation or 95% confidence intervals for all quantitative results to ensure the robustness of their claims.

Action items.

- [science] In Section 4.1 (Image Editing), the user study involves 29 participants but lacks statistical reporting. Please report confidence intervals (e.g., 95% CI) for the mean preference scores and specify the statistical test used to determine significance against baselines (e.g., paired t-test or Wilcoxon signed-rank), including p-values.
- [science] In Section 4.2 (Sudoku), Table 2 reports performance improvements (e.g., −9.0% Exact Accuracy) without indicating statistical significance. Given the ablation study nature, please report standard deviations or confidence intervals across multiple random seeds to confirm these gains are not due to variance.
- [science] In Section 4.3 (Text Reasoning), Table 3 and Table 4 show performance gains on MATH500 and MBPP. The paper does not mention the number of random seeds used for these benchmarks or provide error bars. Please clarify the experimental variance and add statistical significance tests (e.g., bootstrap CIs) to support the claimed improvements.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_writing_quality.md`

The manuscript demonstrates a high standard of academic writing, with clear structure, logical flow, and precise terminology throughout. The abstract effectively summarizes the contributions, and the introduction successfully motivates the problem and solution. The method section is well-organized, with clear definitions of notations and step-by-step explanations of the inference rules and training paradigm.

However, there are a few minor grammatical and stylistic issues that, while not impeding understanding, should be addressed to polish the final version. In Section 3.1, the sentence "In general, at each iteration, model can produce one of three actions" is missing the definite article "the" before "model." Similarly, in Section 4.1, the phrase "The brighter regions in pixel-wise difference heat maps indicates larger pixel change" contains a subject-verb agreement error ("regions" is plural, so "indicates" should be "indicate").

Additionally, some sentences are slightly dense or could be streamlined for better readability. For instance, in Section 3.2, the explanation of the history embedding could be slightly simplified to ensure immediate clarity for readers unfamiliar with the specific rotation mechanics. In Section 4.2, the comparison of decay factors could be phrased more smoothly to avoid the slightly awkward "still improves... but performs worse" construction.

Overall, the writing is strong and professional, with only minor corrections needed to achieve publication-ready quality. The authors have done an excellent job of presenting complex technical concepts in a clear and accessible manner.

Action items.

- [writing] In Section 3.1 (Eq. 1), the text states 'model can produce' but should be 'the model can produce' for grammatical correctness. Additionally, the phrase 'driven entirely by the model's per-position output distribution' is slightly redundant; consider 'driven by the model's per-position output distribution'.
- [writing] In Section 3.2, the sentence 'The current state contributes unrotated, while each past state is added with a lag-dependent rotation R_□ and decay □^|□|' is grammatically sound but slightly dense. Consider splitting or clarifying 'lag-dependent rotation' to ensure the reader immediately grasps the mechanism without re-reading.
- [writing] In Section 4.1, the phrase 'The brighter regions in pixel-wise difference heat maps indicates larger pixel change' contains a subject-verb agreement error: 'regions' (plural) should match 'indicate' (plural), not 'indicates'.
- [writing] In Section 4.2, the sentence 'However, introducing a decay factor alone still improves over the no-history baseline but performs worse than the HR-only variant' is slightly clunky. Consider rephrasing to 'However, introducing a decay factor alone

improves over the no-history baseline but performs worse than the HR-only variant' to remove the redundant 'still' and improve flow.